## §1.3 Lab Assignment 1: Runge's Phenomenon and Chebyshev Nodes

You might expect polynomial interpolation to *converge* as $n \to \infty$. Surprisingly, this is not the case if you take equally-spaced nodes $x_i$. This was shown by Runge in a famous 1901 paper.

### Question 17.

*Use your favorite mathematical or programming software to define Runge's function*

$$f(x) = \frac{1}{1+25x^2}$$

**(a)** *Write code/script to create a list of pairs (data-points) $\{(x_i, f(x_i))\}$ for $(n+1)$ evenly spaced points $x_i$ between $-1$ and 1. For example, for $n = 4$, the points $x_i$ should be 0.5 apart.*

- *Your input should be $n$.*
- *Depending on the programming language you might need separate arrays for $x_i$ values and $f(x_i)$ values for the next part.*

> ***Note:*** *Use* `Table` *for Mathematica,* `numpy.linspace` *for Python.*

**(b)** *Define the $f$−interpolating polynomial $p_n(x)$ corresponding to the lists of nodes.*

> ***Note:*** *Check here for Mathematica and here for Python code for Interpolating Polynomial.*

**(c)** *Plot $f$ and the polynomial $p_i$ on the same graph, for $i = 4, 10, 20$, and describe what you observe.*

**(d)** *Use this to predict how the plot of an analogous $f$−interpolating polynomial $p_{100}$ would compare (to that of $f$).*

**(e)** *Make a list plot of $\log\left(\max_{x\in[-1,1]} |f(x) - p_n(x)|\right)$ along the vertical axis vs. $n$ along the horizontal axis for $n = 0, 1, 2, \ldots, 25$. Use this to discuss the nature of convergence of $p_n(x)$ as $n \to \infty$.*

What you are seeing here is referred to as "Runge phenomenon", named after the German mathematician and physicist Carl David Tolmé Runge who first explored this example.[†] This is not an issue of numerical instability but rather a fatal flaw associated with uniformly spaced interpolation points.

The problem is (largely) coming from the polynomial

$$w(x) = (x - x_0)(x - x_1)(x - x_2)\ldots(x - x_n)$$

that showed up in the expression of $\mathrm{E}_n(x)$ in the last section. Check that the problem is mostly occurring near the ends of the interval, so it would be logical to put more nodes there to reduce the error. A good choice was first proposed by the Russian mathematician Pafnuty Chebyshev (1821-1894). The idea is as follows:

- Draw a semicircle above the closed interval on which you are interpolating ($[-1, 1]$ in this case).

[†] For a proof see `http://math.stackexchange.com/questions/775405/`.

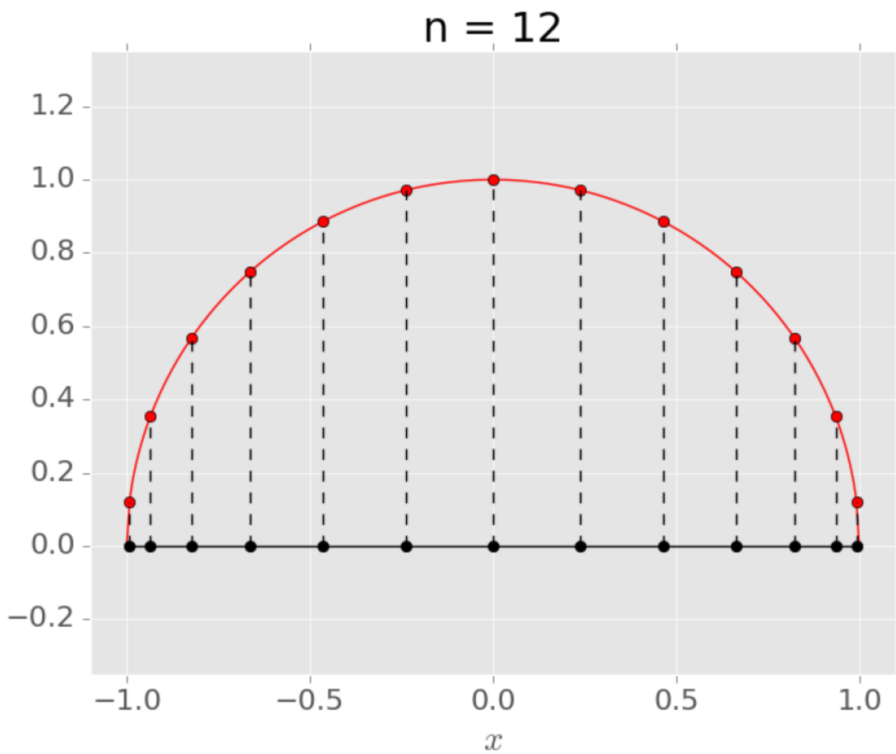


Figure 1.2: Chebyshev Nodes for $n = 12$

- Pick $(n+2)$ equally spaced points, including the end points ($(-1,0)$ and $(1,0)$ in this case), along the semicircle (i.e. same arc length between each point).
- Then choose the midpoints of each of these arcs to get a list of $(n+1)$ points.
- Project the $(n+1)$ points on the semicircle down to the interval. Use these projected points for the interpolation.

## Question 18.

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***(a)*** *Show that the Chebyshev nodes are given by*

$$x_j = \cos\frac{(2j+1)\pi}{2(n+1)}, \qquad j = 0, \ldots, n. \tag{1.8}$$

*Although these values might seem arbitrary, they are actually directly related to the Chebyshev polynomial, which is defined as*

$$T_n(x) := \cos\left[n \arccos(x)\right]$$

***(b)*** *Prove by induction on n that $T_n$ is a degree n polynomial.*

***(c)*** *Prove that the Chebyshev nodes in equation (1.8) are the roots of the Chebyshev polynomial $T_{n+1}(x)$.*

Although we will not prove it here, one can show that the Chebyshev nodes are the best possible choice for minimizing the error coming from $w(x)$.

**Theorem 1.3.6: Chebyshev Interpolation**

*The choice of real numbers*

$$-1 \leq x_0, x_1, x_2, \ldots, x_n \leq 1$$

*that makes the value of*

$$\max_{-1\leq x\leq 1} |w(x)|$$

*as small as possible is:*

$$x_j = \cos\left(\frac{(2j+1)\pi}{2n}\right), \quad j = 0, 1, \ldots, n$$

*Furthermore, the minimum value is* $\frac{1}{2^{n-1}}$.

## Question 19.

*Define the* $f$*−interpolating polynomial* $q_n(x)$ *corresponding to the new set of* $(n+1)$ *Chebyshev interpolation points (note that these are not equally-spaced).*

***(a)*** *Plot* $f$ *and the polynomial* $q_i$ *on the same graph, for* $i = 4, 10, 20$*, and describe what you observe.*

***(b)*** *Make a list plot of* $\log\left(\max_{x\in[-1,1]} |f(x) - q_n(x)|\right)$ *along the vertical axis vs.* $n$ *along the horizontal axis for* $n = 0, 1, 2, \ldots, 25$*. Use this to discuss the nature of convergence for* $q_n(x)$ *as* $n \to \infty$*. Compare what you observe to Runge's phenomenon for equally-spaced interpolation points.*

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